FLEET & SHORE PERSONNEL TRAINING

Incident Investigation

Definitions, causation, evidence, interviewing, reporting and the classification of marine injuries.

ISM Code · OCIMF Marine Injury Reporting Guidelines · REP-U-006 · Q36 Investigator List

COURSE MAP

What we will cover

01 **Definitions and Purpose**
Accident, incident, near miss, loss and non-conformity

02 **The Investigation Team**
Who investigates, composition and training

03 **Steps in an Investigation**
From first report to continual improvement

04 **Causation and Root Cause**
Five categories and the questions to ask

05 **Collecting Evidence**
Physical evidence, records and shipboard data

06 **Interviewing Witnesses**
Technique, questions, do and do not

07 **Analysis and Reporting**
Conclusions, recommendations and follow-up

08 **Reporting to the Company**
Timescales, severity categories and forms

09 **Investigation and Lessons Learnt**
Records, the DPA and fleet distribution

10 **Injury Classification**
OCIMF classes and safety performance measures

SECTION 01

Definitions and Purpose

An incident is an occurrence, condition or situation arising in the course of work that resulted in, or could have resulted in, injuries, illnesses, damage to health or fatalities.

THE PURPOSE OF AN INVESTIGATION

We are looking for facts, not fault

When incidents are investigated, the emphasis should be concentrated on finding the root cause so the event can be prevented from happening again. The purpose is to find facts that can lead to corrective actions, not to find fault. Always look for deeper causes — do not simply record the steps of the event.

Find the root cause

Correct, do not blame

Prevent recurrence

GENERAL TERMINOLOGY

Incident, accident, near miss

Different terms are used in different jurisdictions. For simplicity, this course uses the term incident to cover all of the events below.

Incident

An occurrence, condition or situation arising in the course of work that resulted in, or could have resulted in, injuries, illnesses, damage to health or fatalities. Some use it only for an unexpected event that did not cause injury or damage that time but had the potential to.

Accident

An unplanned event that interrupts the completion of an activity, and that may — or may not — include injury or property damage.

Near miss or dangerous occurrence

An event that could have caused harm but did not. The terms are used interchangeably for an event where, under slightly different circumstances, loss would have followed.

Why we say "incident"

It is argued that the word "accident" implies the event was related to fate or chance. When the root cause is determined, most events turn out to have been predictable and preventable had the right actions been taken — so "incident" is the better word.

COMPANY DEFINITIONS

How the Company defines each term

From the Company procedure on reporting and analysis of non-conformities, accidents and incidents.

Accident

Any incident and/or breach of International, National (Flag State), Class Society, Port State, other authorities, Owner/Managers or Charterer's instructions and procedures which has inadvertently caused a loss or hazard to life, property or environment, and detention of the vessel.

Incident

A dangerous occurrence that leads, or may lead, to any kind of loss. For other accident and incident definitions, see "The Marine Injury Reporting Guidelines" by OCIMF.

Near miss

An event, or sequence of events, that did not result in an injury or accident — without any loss — but which, under slightly different circumstances, could have done so. Also called hazardous occurrence and unsafe acts, or hazardous situations and unsafe conditions.

COMPANY DEFINITIONS

Serious near miss, loss and findings

Serious / significant near miss
An event or sequence of events which did not result in an accident, incident or loss but which, under slightly different conditions, may have caused loss of life, loss of the vessel or extreme environmental impact.

Loss
Any hazard to life, property, environment or negative business impact. Losses may be incurred immediately after an incident or only after a certain period.

Non-conformity
Any item found by any means — Port State Control, Flag State Control, Company or ship personnel — to be not in compliance with, or in breach of, any specified requirement of the ISM, ISPS and MLC Codes, ISO and OHSAS standards or the Company Management System.

Observation
A statement of fact made during a safety management audit and substantiated by objective evidence. Findings during SIRE, CDI or company inspections that do not comply with the requirements defined on the check lists are also defined as observations.

Deficiency
Any equipment or technical system found by any means, including by a third party, the Company or ship personnel, to be not in compliance with any specified requirement.

THE ESCALATION CHAIN

The same defect, four different outcomes

A worn lifting sling illustrates how one unaddressed condition escalates. Only the timing of intervention differs between a report and a fatality.

1. Unsafe condition

The lifting gear is corroded and worn. The defect exists but nothing has happened yet — this is the cheapest point to intervene.

2. Unsafe act

The gear is used anyway. The condition has now been combined with a decision, and the risk is live.

3. Near miss

The sling parts and the load falls clear of the crew. No injury, no loss — but the outcome was decided by chance, not by control.

4. Accident

The same failure, a metre to the left. Injury, damage, delay and investigation. Nothing about the defect changed.

WHY WE DO IT

Reasons to investigate, and where the approach applies

Reasons to investigate an incident

- Most importantly, to find out the cause of incidents and to prevent similar incidents in the future
- To fulfil any legal requirements
- To determine the cost of an incident
- To determine compliance with applicable regulations, for example occupational health and safety or criminal law
- To process workers' compensation claims

Where the same principles apply

- The same principles apply to an inquiry into a minor incident and to the formal investigation of a serious event
- The steps can be used to investigate any situation — including one where no incident has occurred yet
- Used that way, the process becomes a means of preventing an incident rather than reacting to one

Losses that are not obvious

- Interruption or suspension of an operation
- Cease of work
- Voyage cancellations or suspensions
- Time loss
- Consequences of all of the above

SECTION 02

The Investigation Team

02

An investigation is only as credible as the people conducting it. Independence, competence and training all have to be demonstrable.

What an investigator needs to be able to do

Ideally an investigation is conducted by a person or group who are experienced, knowledgeable and able to work with people.

- Experienced in incident causation models
- Experienced in investigative techniques
- Knowledgeable of any legal or organisational requirements
- Knowledgeable in occupational health and safety fundamentals
- Knowledgeable in the work processes, procedures, persons and industrial relations environment for that particular situation
- Able to use interview and other person-to-person techniques effectively, such as mediation or conflict resolution
- Knowledgeable of requirements for documents, records and data collection
- Able to analyse the data gathered to determine findings and reach recommendations

Jurisdictional requirements

Some jurisdictions require that the investigation be conducted jointly, with both management and labour represented, or that the investigators be knowledgeable about the work processes involved. Where a serious injury or fatality has occurred, other authorities may have jurisdiction — police or OH&S inspectors — and may take control of the scene. The organisation should establish, implement and maintain a procedure to coordinate incident management with the authority having jurisdiction.

Who can be a member of the team

Employees with knowledge of the work

The people who understand how the task is actually performed.

Supervisor of the area or work

Closest to the conditions and able to act immediately.

Safety officer

Brings the health and safety framework to the inquiry.

Health and safety committee

Provides continuity with the wider safety programme.

Union representative, if applicable

Ensures the workforce view is represented.

Employees experienced in investigations

Bring method and discipline to the evidence gathering.

"Outside" experts

Called in for specialist analysis of equipment or materials.

Local government or police representative

Where the authority having jurisdiction is involved.

A COMMON QUESTION

Should the immediate supervisor be on the team?

The advantage

- This person is likely to know most about the work and the persons involved
- And most about the conditions prevailing at the time
- The supervisor can usually take immediate remedial action

The counter-argument

- There may be an attempt to gloss over the supervisor's own shortcomings in the incident

How the risk is removed

- The situation should not arise if the incident is investigated by a team of people
- And if the worker representatives and the investigation team members review all findings and recommendations thoroughly

INDEPENDENCE

Who may be appointed, and on what basis

At least two qualified and trained people

At least two people who are experienced and trained in accident and incident investigation are dedicated according to the type of incident.

Not connected with the event

Investigators from ship and office who are not connected with the incident, accident or serious / significant near miss attend, to provide an independent and fair-minded investigation.

A systematic method

Investigators should use a systematic methodology and / or tool — the BP List of Causes Charts — to determine root causes. The same tool applies to each near miss report.

Third parties for independence

Third-party contractors who are appropriately qualified may be employed whenever deemed necessary to maintain independence.

Q36 Investigator List

The Company maintains a file — Q36 Investigator List — containing a list of all investigators, including the vessel personnel who can conduct or assist with an investigation.

Investigation team by severity category

As per the severity categories described in the Risk Matrix and Severity Table.

	Category 1-2 / Negligible - Minor	Category 3 / Significant	Category 4-5 / Critical - Catastrophic
Who investigates	Master, Chief Engineer, Chief Officer and crew who are not part of the incident and are trained may be invited by the team. The Superintendent will support and guide them during the investigation.	Vessel Superintendents to report the causes and the action plan to the DPA.	Investigation team to be appointed by the Fleet Manager. Outsourced investigation team may be invited if required by the Company, given the importance or severity of the incident.
Report deadline	The report is to be finalised within one week of the incident.	The report is to be finalised within one week of the incident.	The report is to be finalised within 3 months where circumstances permit.
Attendance from ashore	Not normally required.	For cases categorised 3 and above, a Company employee is to be sent from the office on board the vessel.	For cases categorised 3 and above, a Company employee is to be sent from the office on board the vessel.

If the same incident or accident happens several times a year in the Company, the investigation team is to be appointed by the Fleet Manager regardless of the category. If potential significant outcomes are identified for significant or critical near misses and minor incidents, the Company considers investigation by a shore team.

Investigation team by type of incident

	Navigation, personal injury, cargo / ballast and mooring, oil spill, security	Equipment failure, engine breakdown, bunkering	Other types of incident
Types covered	Navigation related incidents · Personal injury, accident or illness · Cargo / ballast and mooring operation related incident · Oil spill incident · Security related incident	Equipment failure and engine breakdown incidents · Bunkering related incidents	Other type of incidents
Investigators	HSEQ Superintendents and / or Master and / or Chief Officer	Technical Superintendents and / or Chief Engineer and / or 2nd Engineer	Any suitable investigator

Third-party contractors who are appropriately qualified may be employed whenever deemed necessary to maintain independence.

Training of the investigation team

Investigation team members are appropriately trained and qualified to conduct the investigation and analysis. See the M6 Training Manual for more detail.

Accepted training routes for vessel crew

An industry recognised training programme · appropriate in-house training by an accredited trainer · appropriate computer-based training.

Shore-based management teams

Shore based management teams receive external training from industry recognised training providers.

Practical experience before leading

Trained personnel are invited or nominated onto investigations to gain practical experience and must participate in at least one investigation, practising the relevant skills, before leading one. The DPA considers this before assigning a person to an investigation.

Officers joining the managed vessels

Officers employed on board the managed vessels are trained on accident and incident investigation in the office before joining, or by the Superintendents during ship visits where practicable, and are then nominated onto investigations for further experience.

SECTION 03

03

Steps in an Investigation

As little time as possible should be lost between the moment of an incident and the beginning of the investigation.

BEFORE ANYTHING ELSE

The first response

Two things happen before any investigative activity begins.

1. Report the occurrence

Report the incident occurrence to a designated person within the organisation.

2. Care for the injured

Provide first aid and medical care to the injured person or persons.

3. Prevent further harm

Prevent further injuries or damage before any evidence gathering starts.

Five general steps for the investigation team

1. **Scene management and scene assessment**
   Secure the scene and make sure it is safe for investigators to do their job.

2. **Witness management**
   Provide support, limit interaction between witnesses, and interview them.

3. **Investigate the incident and collect data**
   Gather physical evidence, witness accounts and documentary records.

4. **Analyse the data and identify the root causes**
   Work back from the moment of the incident and list all possible causes at each step.

5. **Report the findings and recommendations**
   Set out what happened, why, and what should change.

AND THEN THE ORGANISATION

Closing the loop after the report

Develop a plan for corrective action

Turn the recommendations into a plan with owners and dates.

Implement the plan

The department head or Master concerned implements the corrective action.

Evaluate the effectiveness of the corrective action

The effectiveness of the corrective action is to be verified, not assumed.

Make changes for continual improvement

Feed what has been learned back into procedures, training and design.

Lose no time

As little time as possible should be lost between the moment of an incident and the beginning of the investigation. In this way one is most likely to observe the conditions as they were at the time, prevent disturbance of evidence, and identify witnesses. The tools the team may need — pencil, paper, camera or recording device, tape measure — should be immediately available so that no time is wasted.

SECTION 04

04

Causation and Root Cause

Even in the most seemingly straightforward incidents, seldom — if ever — is there only a single cause.

A shallow finding closes off the useful questions

An investigator who believes incidents are caused by unsafe conditions will uncover conditions; one who believes they are caused by unsafe acts will find human errors. It is necessary to examine all underlying factors in the chain of events.

What a shallow inquiry concludes

The investigation stops here

- The incident was due to worker carelessness
- No further questions are asked
- Nothing correctable has been identified

What it failed to ask

Each answer opens a correctable condition

- Was the worker distracted? If yes, why was the worker distracted?
- Was a safe work procedure being followed? If not, why not?
- Were safety devices in order? If not, why not?
- Was the worker trained? If not, why not?

Five categories of cause

Many models of causation have been proposed, from Heinrich's domino theory to the sophisticated Management Oversight and Risk Tree (MORT). This simple model groups the causes of any incident into five categories. When it is used, possible causes in every category should be investigated.

Task

The actual work procedure being used at the time of the incident.

Material

The equipment, products and raw materials in use.

Environment

The physical work environment, especially sudden changes to it.

Personnel

The physical and mental condition of those directly involved.

Management

Supervision, systems and organisational factors.

CATEGORY 1

Task — the work procedure in use

Here the actual work procedure being used at the time of the incident is explored. These are sample questions only; no attempt has been made to develop a comprehensive checklist.

- Was a safe work procedure used?
- Had conditions changed to make the normal procedure unsafe?
- Were the appropriate tools and materials available?
- Were they used?
- Were safety devices working properly?
- Was lockout used when necessary?

The follow-up question

For most of these questions, an important follow-up question is "If not, why not?" Each time the answer reveals an unsafe condition, the investigator must go on to ask why that situation was allowed to exist. Without it, the inquiry records the event rather than explaining it.

CATEGORIES 2 AND 3

Material and work environment

For the work environment, the situation at the time of the incident is what matters — not what the "usual" conditions were.

Material — equipment and products

- Was there an equipment failure? What caused it to fail?
- Was the machinery poorly designed?
- Were hazardous products involved? Were they clearly identified?
- Was a less hazardous alternative product possible and available?
- Was the raw material substandard in some way?
- Should personal protective equipment (PPE) have been used? Was the PPE used?
- Were users of PPE properly educated and trained?

Work environment — conditions at the time

- What were the weather conditions?
- Was poor housekeeping a problem?
- Was it too hot or too cold?
- Was noise a problem?
- Was there adequate light?
- Were toxic or hazardous gases, dusts or fumes present?

CATEGORY 4

Personnel — physical and mental condition

The purpose of investigating is not to establish blame, but the inquiry will not be complete unless personal characteristics and psychosocial factors are considered. Some factors remain essentially constant; others vary from day to day.

- Did the worker follow the safe operating procedures?
- Were workers experienced in the work being done?
- Had they been adequately educated and trained?
- Can they physically do the work?
- What was the status of their health?
- Were they tired? Was fatigue or shiftwork an issue?
- Were they under stress, at work or personally?
- Was there pressure to complete tasks under a deadline, or to by-pass safety procedures?

Condition, not character

The physical and mental condition of those individuals directly involved must be explored, together with the psychosocial environment they were working within. This is an examination of the conditions a person was working under — not an assessment of the person.

CATEGORY 5

Management — the organisational factors

Management holds the legal responsibility for the safety of the workplace. Failures of management systems are often found to be direct or indirect causes.

Procedures, training and supervision

- Were safety rules or safe work procedures communicated to and understood by all employees?
- Were written procedures and orientation available?
- Were the safe work procedures being enforced?
- Was there adequate supervision?
- Were workers educated and trained to do the work?

Hazards, maintenance and reporting

- Had hazards and risks been previously identified and assessed?
- Had procedures been developed to eliminate the hazards or control the risks?
- Were unsafe conditions corrected?
- Was regular maintenance of equipment carried out?
- Were regular safety inspections carried out?
- Had the condition or concern been reported beforehand? Was action taken?

THE POINT OF THE FIVE CATEGORIES

Never accept a single cause

This model provides a guide for uncovering all possible causes and reduces the likelihood of looking at facts in isolation. Some investigators may prefer to place a question in a different category; the categories are not important, as long as each question is asked. There is considerable overlap between them — and that overlap reflects real life.

Ask every question

Then ask why not

Expect overlap

SECTION 05

Collecting Evidence

05

Physical evidence is the most non-controversial information available — and the most quickly lost. It should be recorded first.

Four steps, each with its own pitfalls

The steps are simple, but each one can go wrong. An open mind is necessary: preconceived notions lead down wrong paths and leave significant facts uncovered.

1. Gather the data

Examine the site for a quick overview, take steps to preserve evidence, and identify all witnesses.

2. Analyse it

Consider all possible causes. Making notes of ideas as they occur is good practice.

3. Determine the findings

Conclusions should not be made until all the data has been gathered.

4. Make recommendations

Well-considered measures designed to prevent recurrence of similar incidents.

What to record before anything is moved

Physical evidence is subject to rapid change or obliteration, so it should be the first to be recorded. Based on your knowledge of the work process, check items such as these.

- Positions of injured workers
- Equipment being used and products being used
- Safety devices in use and the position of appropriate guards
- Position of controls of machinery
- Damage to equipment
- Housekeeping of the area
- Weather conditions and lighting levels
- Noise levels and time of day

Photographs, sketches and samples

- In some jurisdictions an incident site must not be disturbed without approval from the coroner, inspector or police
- Take photographs before anything is moved — of the general area and of specific items; a later study may reveal what was missed
- Sketches based on measurements taken help in later analysis and clarify the written report
- Broken equipment, debris and samples of materials may be removed for analysis by appropriate experts
- Even where photographs are taken, written notes about the location of these items at the scene should be prepared

Other information to be studied

Any relevant information should be studied to see what might have happened, and what changes might be recommended to prevent recurrence of similar incidents.

Technical data sheets

Manufacturer data on the equipment and products in use.

Health and safety committee minutes

Concerns previously raised and how they were handled.

Inspection reports

Findings from internal and external inspections.

Company policies

What the organisation required to be done.

Maintenance reports

Work carried out, deferred and outstanding.

Past incident reports

Whether this has happened, or nearly happened, before.

Safe-work procedures

The procedure that should have been in use.

Training reports

What the people involved had been trained to do.

ON BOARD

Additional shipboard evidence to consider

Beyond the general sources, investigators should consider the shipboard information available to them.

D&A testing
Whenever an incident or accident is observed, an alcohol test must be made on all personnel involved and the results reported in the relevant form.

Review of work and rest hours
Records for the personnel involved in the period leading up to the event.

Witness statements
Written statements from officers, ratings and any known witnesses.

Photographic evidence and CCTV
Images of the damaged areas and any recorded coverage of the scene.

VDR, ECDIS and VTS data
VDR and / or ECDIS data, and evidence from vessel traffic services.

Records, log books and forms
Maintenance history including modifications of the equipment; log books, permits, check lists and the relevant records and forms.

SECTION 06

Interviewing Witnesses

06

Witnesses may be under severe emotional stress, or afraid to be completely open for fear of recrimination. Interviewing them is probably the hardest task facing an investigator.

Before the interview starts

Make every effort to interview

There may be occasions when you are unable to, but every effort should be made. In some situations witnesses are the primary source of information, because you may be asked to investigate without being able to examine the scene immediately after the event.

Keep witnesses apart

Witnesses should be kept apart and interviewed as soon as possible after the incident. If they discuss the event among themselves, individual perceptions may be lost in the normal process of accepting a consensus view where doubt exists about the facts.

Interview alone, not in a group

Each witness is interviewed individually so that their own account is obtained rather than a shared one.

Choose the location deliberately

At the scene it is easier to establish the positions of each person involved and obtain a description of events. A quiet office has fewer distractions. The decision depends on the nature of the incident and the mental state of the witness.

Conducting the interview

The purpose of the interview is to establish an understanding with the witness and to obtain his or her own words describing the event.

DO

- Put the witness, who is probably upset, at ease
- Emphasise the real reason for the investigation — to determine what happened and why
- Let the witness talk, and listen
- Confirm that you have the statement correct
- Try to sense any underlying feelings of the witness
- Make short notes, or ask someone else on the team to take them during the interview
- Ask if it is okay to record the interview, if you are doing so
- Close on a positive note

DON'T

- Intimidate the witness
- Interrupt
- Prompt
- Ask leading questions
- Show your own emotions
- Jump to conclusions

THE QUESTIONS

Seven questions to ask every time

Ask open-ended questions that cannot be answered simply "yes" or "no". The actual questions will vary with each incident, but these should be asked each time.

- Where were you at the time of the incident?
- What were you doing at the time?
- What did you see, and what did you hear?
- What were the work environment conditions — weather, light, noise and so on — at the time?
- What was, or were, the injured worker or workers doing at the time?
- In your opinion, what caused the incident?
- How might similar incidents be prevented in the future?

Assess the accuracy

Asking questions is a straightforward approach to establishing what happened. But care must be taken to assess the accuracy of any statements made in the interviews. Corroborate each account against the physical evidence and the other statements before treating it as fact.

ONE FURTHER TECHNIQUE

Re-enactment, used with care

Why it is used

Another technique sometimes used to determine the sequence of events is to re-enact or replay them as they happened.

How it is done

A witness — usually the injured worker — is asked to re-enact in slow motion the actions that happened before the incident.

The condition attached

Care must be taken so that further injury or damage does not occur. If the re-enactment cannot be done safely, it is not done.

07

SECTION 07

Analysis and Reporting

Data gathering represents only the first half of the objective. Now comes the key question — why did it happen?

MAKING THE ANALYSIS

From what happened to why it happened

Keep an open mind

At this stage most of the facts about what happened and how it happened should be known. Keep an open mind to all possibilities and look for all pertinent facts.

Fill the gaps

There may still be gaps in your understanding of the sequence of events. You may need to re-interview some witnesses or look for other data to fill them.

Write the account backwards

When the analysis is complete, write down a step-by-step account of what happened — the team's conclusions — working back from the moment of the incident and listing all possible causes at each step. This is not extra work: it is a draft for part of the final report.

Test every conclusion

Check each conclusion to see whether it is supported by evidence; whether that evidence is direct — physical or documentary — or based on eyewitness accounts; or whether it is based on assumption. The resulting list serves as a final check on discrepancies that should be explained.

Specific beats general, every time

The most important final step is to come up with a set of well-considered recommendations designed to prevent recurrences of similar incidents. Resist the temptation to make only general recommendations to save time and effort.

Recommendations should

- Be specific
- Be constructive
- Identify root causes
- Identify contributing factors

Too general to act on

A blind corner contributed to the incident

- "Eliminate blind corners"
- Nobody owns it, nothing is measurable, and the corner stays as it is

Specific and general together

The same finding, written usefully

- Install mirrors at the northwest corner of building X — specific to this incident
- Install mirrors at blind corners where required throughout the worksite — general

A RULE WITH NO EXCEPTIONS

Never recommend discipline

Never make recommendations about disciplining a person or persons who may have been at fault. This action would not only be counter to the real purpose of the investigation, it would jeopardise the chances of a free flow of information in future investigations. Any disciplinary steps belong within the normal personnel procedures.

Remedy the situation | **Not the individual** | **Protect future reporting**

Name the failing, do not punish the person

A difficulty that has bothered many investigators is the idea that one does not want to lay blame.

Point it out

- Where a thorough worksite investigation reveals that some person or persons among management, supervisors and workers were apparently at fault, that fact should be pointed out
- The intention is to remedy the situation, not to discipline an individual
- Failing to point out human failings downgrades the quality of the investigation
- It also allows future incidents to happen from similar causes, because they have not been addressed

When causes remain uncertain

- In the unlikely event that the causes could not be determined with complete certainty, you have probably still uncovered weaknesses within the process or the management system
- It is appropriate that recommendations be made to correct those deficiencies
- Any disciplinary steps should be done within the normal personnel procedures

The measure of a good report is quality, not quantity

Write for a reader who was not there

The prepared draft of the sequence of events can now be used to describe what happened. Readers do not have your intimate knowledge of the incident, so include all relevant details, including photographs and diagrams.

Show the basis of every statement

Identify clearly where evidence is based on certain facts, on witness accounts, or on the team's assumptions. If doubt exists about any particular part of the event, say so. State the reasons for your conclusions, then give your recommendations.

Leave out what is not needed

Do not include extra material that is not required for a full understanding of the incident and its causes — irrelevant photographs, or parts of the investigation that led nowhere.

Communicate the findings in context

Always communicate findings and recommendations with workers, supervisors and management. Present the information in context so everyone understands how the incident occurred and the actions needed to prevent it happening again.

Forms and checklists

Some organisations may use pre-determined forms or checklists. Use these documents with caution, as they may be limiting in some cases. Always provide all of the information needed to help others understand the causes of the event, and why the recommendations are important.

How follow-up should be done

Management is responsible for acting on the recommendations in the investigation report. The health and safety committee or representative, if present, can monitor the progress of these actions.

1. **Respond to the recommendations**
   Explain what can and cannot be done — and why, or why not.

2. **Develop a timetable for corrective actions**
   Each action given an owner and a date.

3. **Monitor that the scheduled actions have been completed**
   Completion is verified, not assumed.

4. **Check the condition of the injured worker or workers**
   The people affected remain part of the follow-up.

5. **Educate and train other workers at risk**
   Extend the learning beyond those directly involved.

6. **Re-orient workers on their return to work**
   Returning personnel are brought back up to date before resuming duties.

SECTION 08

Reporting to the Company

08

All near misses, deficiencies, observations and non-conformities occurring on board, as well as suggestions for improvement, must be reported to the Company.

What must be reported, and why

Everything is reported and recorded

All near misses, deficiencies, observations and non-conformities occurred on board, as well as suggestions for improvements, must be reported to the Company. All reports are kept in the MPM HSEQ Module.

It is Company policy to motivate reporting

It is the Company's policy to motivate both shore and shipboard personnel to report any unplanned event involving Company personnel or assets which threatened but did not cause injury or loss, since proper preventive action is proven to decrease accidents and incidents.

All crew members are encouraged to report

All crew members are encouraged to report all near misses and incidents, and so take part in the task of improving the safety of all kinds of operations within the fleet vessels.

The safety culture addresses hazards

The Company safety culture encourages all personnel to identify, report and, where applicable, address hazards. Any identified hazards should be clearly addressed, and hazards that cannot be rectified by vessel personnel should be reported to Company management so that remedial action can be taken.

NOTIFICATION

The reporting sequence after an event

Incidents, accidents and serious or significant near misses should be promptly reported to the Company as soon as possible after the event, by the most appropriate means — preferably by phone.

1. **Telephone the Company immediately**
   Prompt notification by the most appropriate means, preferably by phone, as soon as possible after the event.

2. **Submit a preliminary report**
   A preliminary report providing the first available information is submitted by the relevant department head, using the Incident or Accident Report Forms.

3. **Send the initial report within 24 hours**
   The initial report is sent as soon as possible after the event, but not later than 24 hours from occurrence, by completing Section A of the incident report.

4. **Comply with flag and charterer requirements**
   National and flag administration requirements for reporting marine casualties must be complied with. Charterer's instructions are followed for reporting incidents, accidents and serious or significant near misses to charterers.

Classification of near miss reports by severity category

A systematic methodology and / or tool — the BP List of Causes Charts — should be in use to determine root causes for each near miss report.

Consequence	Category 1 / Negligible	Category 2 / Minor	Category 3 / Significant	Category 4 / Serious	Category 5 / Catastrophic
People	Injury is not credible (First Aid Case)	Only a minor injury is credible (Medical Treatment Injury)	A single serious injury is credible (RWC)	Fatality / multiple serious injury is credible (DAFWC)	Multiple fatality is credible (Fatality)
Assets	Less than $1,000	Less than $10,000 but greater than $1,000	Greater than $10,000 but less than $99,999	Vessel disabled or >= $100,000	Explosion / collision / grounding or > $100,000
Operational and financial loss	No missed voyage	<$10K additional work; no missed voyage	Longer operational and financial loss $10K-$100K	Vessel disabled or >= $100,000	Greater than $1M
Environment	Loss of containment <1 Bbl, not on deck or to environment	Environmental impact not existent	Local environmental impact	Regional environmental impact	Major spill >= 100 Bbls, or uncontrolled gas release >10 tonnes or volumetric equivalent
Reputation	Zero impact	Limited impact	Considerable impact	Major national impact	Major international impact

EXTERNAL NOTIFICATION

Sharing incidents outside the Company

Incidents, accidents and serious or significant near misses of Category 3 (Significant) or higher are reported to Oil Major Vetting departments.

E-mail is the primary method

E-mail is used as the primary method for both initial notification and the final report to Oil Major Vetting departments for Category 3 and above.

Lower severity is shared too

The Company also shares lower level severity incidents and significant near misses which have a potential significant outcome.

OCIMF SIRE database

The OCIMF SIRE database is considered as an additional option for data sharing related to incidents.

Within 15 days

In any case, related oil major vetting departments will be contacted within 15 days to find out whether any further information is required by them, where appropriate.

Getting the warning to the rest of the fleet

Urgent safety related information, including incidents and serious or significant near misses, must be distributed rapidly to fleet vessels as a SAFETY ALERT NOTICE by e-mail, and verification should be requested.

Not later than 24 hours

Distribution is as soon as possible after the event, but not later than 24 hours from occurrence.

Immediate action on board

Where an incident has occurred and the Company identifies immediate issues of concern to other fleet vessels, immediate investigative and preventative actions must be addressed on board. Proactive measures should minimise the exposure of personnel, environment and property to a hazardous situation.

The DPA is responsible

The DPA is responsible for ensuring this is completed. During weekends and holidays, the DPA coordinates with the duty ERT staff and related managers to complete it.

Superintendents verify

Superintendents are responsible for verification that the actions have been completed on each vessel as per the Lessons Learnt, to avoid repeated incidents. The DPA follows completion with the related managers.

What the report must address

Injury or illness to vessel staff resulting in lost time from work and all other injuries, and vessel incidents involving structural damage to the ship or damage to property other than the ship, must be reported as per REP-U-006.

Personal injury, accident or illness

- Seaman's name and position, identification number and nationality
- Date, place and time of occurrence of the injury
- Date and time when the illness was reported
- A description of the circumstances involving the injury, including environmental conditions
- Circumstances involving the illness
- Statements from other vessel officers or ratings relating to the incident, including the name of each person giving a statement
- Names of any known witnesses to the incident
- Description of radio medical advice received, which must also be entered in the Deck / Medical Log
- Description of any medical action taken

Vessel incidents

- Date, place and time of occurrence
- Circumstances involved, including mitigating factors such as adverse weather, current, visibility or pilot involvement
- Description of damage to the Company vessel
- Description of damage to other property, if applicable
- Description of oil spill extent and emergency response — SOPEP / SMPEP activation — if applicable
- Name of owners or representatives of other property, if applicable and available
- Statements from other vessel officers or ratings relating to the accident, including the name of each person giving a statement
- Names of any known witnesses to the accident
- Number of known injured personnel, if any
- Involvement of local or port state authorities, if any

SECTION 09

Investigation and Lessons Learnt

Upon receipt of notification, the DPA must arrange the investigation team for further investigation and analysis, sufficiently detailed to accurately establish the root causes.

09

What must be recorded, and where it goes

- These items should be recorded in the Deck Logbook.
- A copy of the log, and a copy of any incident form required by the customer or authorities, shall be forwarded by mail to the Company for further handling.
- Any medication dispensed from the vessel's infirmary or medical supply must be recorded.
- Photographs of damaged areas should be taken by a vessel officer to support the vessel's position and better describe the extent of damage, and should accompany the accident report to the DPA where possible.
- As with injury and illness reports, only the facts are to be recorded. Neither a crew member nor a representative of the Company is to admit fault in an accident, as this must be determined by qualified investigators.
- Whenever an incident or accident is observed, an alcohol test must be made on all personnel involved and the results reported in the relevant form.

Incident free days

Incident free days for each fleet vessel will be kept, and this will be posted in the vessel's mess rooms to motivate the crew members. The Company ensures all incidents, accidents and serious or significant near misses are promptly investigated. The investigating team could comprise office personnel, vessel crew and / or third-party contractors as per "Composition of Investigation Team".

SCOPE

What an investigation is conducted on

The Master or the relevant Head of Department, immediately after recording the incident, initiates an investigation in coordination with the DPA. Investigation is conducted on the following, but not limited to.

Fire

Any outbreak of fire on board, whatever its extent.

Oil spill and pollution

Oil spill and pollution events of any kind.

Cargo damage

Damage to cargo in the Company's care.

Collision and grounding

Collision with a ship, wharf, pier or buoy, and grounding.

Accident involving serious injury

Any accident that results in serious injury to a person.

Security breaches

Breaches of the ship security plan and security arrangements.

Near misses and minor injuries with potential

Near misses of significant severity or above, or minor injuries which could potentially have had more severe results had circumstances changed slightly.

Damage to plant and machinery

Damage to plant, equipment, boiler or machinery which compromises safety or seaworthiness.

METHOD

The steps taken during the investigation

1. **Inspect the scene and the equipment involved**
   Establish the physical facts before they change.

2. **Interview personnel and witnesses**
   Including the person directly supervising the work.

3. **Examine any applicable work instructions and procedures**
   What the procedure required, as against what was done.

4. **Record specific instructions that were given on this occasion**
   Instructions given for this particular job, not the standing ones.

5. **Review performance, knowledge and personal characteristics of the employees involved**
   The condition and competence of the people doing the work.

6. **Determine whether other employees were performing similar tasks in the same way**
   Establishes whether this was an individual deviation or common practice.

What the finished report must include, and what the DPA does with it

Having gathered all the facts, the relevant sections of the incident report forms must be completed. The finished investigation report is sent to the DPA.

Incident, accident and serious near miss reports must include

- A description of the event
- Root cause identification
- Witness statement
- A description of probable causes
- A description of consequences with respect to harm to people and damage to property, environment and loss of operational ability
- The description of any corrective or preventive action taken

After the DPA has evaluated it

- The DPA determines with whom else corrective or preventive actions should be discussed, and holds informal meetings with them to complete the analysis
- This includes analysing the identified root causes and drawing conclusions from the investigation
- The lessons learnt are effectively applied throughout the Company to avoid repeat incidents, as per 9.3.5
- The Department Head or Master concerned implements the corrective action, and its effectiveness is to be verified
- Defined preventive action is processed, followed up and verified similarly to corrective action

The DPA is responsible for three things

Defining the root cause and the actions

Defining the root cause, corrective and preventive actions in consultation with relevant shore-based Managers, Masters or the Company's Management Team, as appropriate.

Accurate identification

Ensuring that the root causes and the factors contributing to an incident or significant near miss, and the analysis, are accurately identified.

Following up to effectiveness

Following up corrective actions with related managers to ensure that they have been implemented and are effective.

Who else is notified

- After assessing the incident report, the DPA defines which authorities, makers, subcontractors or other parties need to be notified where the determined root cause addresses possible breaches of Company and legislative requirements
- The relevant shore-based managers or Masters are responsible for the implementation of the defined corrective and preventive actions

What every employee must report, and the HR check

All employees of the Company should always report the following to their department head, through the appropriate forms or through MPM as appropriate.

Reportable to your department head

- Accidents
- Incidents
- Non-conformity
- Customer complaints
- Near misses
- Suggestions for improvement

The HR manager's review

- The HR manager reviews all medical matter records of the crew members before filing them in their personal files
- Where a work related or non-work related accident is identified, this is reported to the DPA immediately
- This ensures it is further investigated as per Company requirements

Distribution, records and benchmarking

Distribution of the investigation report

Upon completion of investigation and analysis, the reports are distributed to all parties involved — such as Flag State and Coastal Authorities, Port State, Classification Society and the Qualified Individual, if applicable.

Lessons learnt go out to the industry

Lessons learnt and incident analysis data are recorded by shore responsible personnel and must be delivered to the entire fleet and to industry groups — classification societies, professional institutes, industry associations, equipment manufacturers and major oil vetting departments, as appropriate.

Records of all incident types

The Company keeps records of injuries to personnel, navigational incidents, mooring incidents, oil spills, machinery failure and incidents related to cargo and ballast transfer. All records are followed within the ISM system and reviewed during management review meetings.

Benchmarking

The Company uses the shared data for benchmarking purposes. Results of benchmarking are used to drive safety performance whenever applicable.

Reviewed at shipboard safety meetings

Incident, accident and near miss reports are promulgated to the fleet vessels via the MPM HSEQ Module and are required to be reviewed during shipboard safety meetings.

SECTION 10

10

Injury Classification

Marine Injury Evaluation and Reporting, following the OCIMF guidelines — consistent classification is what makes safety performance comparable.

Objective and scope

The primary purpose

An increased understanding and awareness of personnel safety, through the efficient and accurate reporting and recording of accidents. The computerised PMS collects data through the accident and incident reports and monitors it up to date automatically.

A consistent method

To provide a consistent method for collecting, classifying, reporting and communicating data on all injuries occurring on board, the monitoring of which provides a measure of the effectiveness of safety management systems.

Comparison and coordination

To facilitate the comparison of safety performance with others, and to coordinate practices and policies with the objective of reducing the frequency of injuries to seafarers.

Scope and definition of Company

The collection of data for injuries occurring to seafarers serving on board tankers. "Company" means the vessel's operating and / or owning Company.

What it does not address

This evaluation and reporting does not address the subject of occupational illnesses, or deaths from natural causes.

Incident, and what is excluded from it

Incident

- An uncontrolled or unplanned event, or sequence of events, that results in a fatality or injury to a seafarer on board ship, or whilst ashore on Company business

This excludes

- Suicide or attempted suicide
- Criminal or terrorist activity
- A deliberate act on the part of another individual
- Incidents which occur off the ship but where the consequences appear on board at some later time

Work injury

- Any sign or symptom of physical damage or impairment to any part of the body directly resulting from an incident
- This applies regardless of the length of time between the incident and the appearance of the injury

THE CLASSES

How a case is classified

Each case is placed in one class. The classes determine which performance indicators the case counts towards.

Fatality

A death directly resulting from a work injury, regardless of the length of time between the injury and the death.

Permanent Total Disability (PTD)

Any work injury which incapacitates an employee permanently and results in termination of employment on medical grounds — for example loss of limbs, permanent brain damage or loss of sight — and precludes the individual from working either at sea or ashore.

Permanent Partial Disability (PPD)

Any work injury which results in the complete or permanent loss of use of any member or part of the body, or any pre-existing disability of the injured member or impaired body function, that partially restricts or limits the employee's basis to work on a permanent basis at sea. Such an individual could be employed ashore but not at sea, in line with industry guidelines.

Lost Workday Case (LWC)

An injury which results in an individual being unable to carry out any of his duties, or to return to work on a scheduled work shift on the day following the injury, unless caused by delays in getting medical treatment ashore. An injury is classified as an LWC if the individual is discharged from the ship for medical treatment.

Restricted Work Case (RWC)

An injury which results in an individual being unable to perform all normally assigned work functions during a scheduled work shift, or being assigned to another job on a temporary or permanent basis on the day following the injury.

Medical Treatment Case (MTC)

Any work-related loss of consciousness, injury or illness requiring more than first aid treatment by a physician, dentist, surgeon or registered medical personnel — or, at sea with no physician on board, treatment that could be considered as being in the province of a physician.

First Aid Case (FAC)

Any one-time treatment and subsequent observation of minor injuries such as bruises, scratches, cuts, burns or splinters. The first aid may or may not be administered by a physician or registered professional.

Near Miss

An event or sequence of events which did not result in an injury but which, under slightly different conditions, could have done so.

What counts as less than normal assigned work functions

The following come into the category of "less than normal assigned work functions" for the purpose of classifying a Restricted Work Case.

Reduced schedule

- Performing all duties or normal assigned work functions, but at less than a full-time schedule

Limited duties

- Performing limited duties at the normally assigned job at a full-time schedule

Transfer

- Transfer to other duties

MEDICAL TREATMENT CASE

What is inside and outside an MTC

MTCs INCLUDE

- Injuries which result in loss of consciousness, even if the individual resumes work after regaining consciousness — this does not cover loss of consciousness due to ill health
- Sutures for non-cosmetic purposes
- Use of casts, splints or other means of immobilisation
- Any general surgical treatment
- Removal of embedded objects from the eye by surgical means
- Use of other than non-prescriptive drugs or medications
- Use of a series of compresses for treatment of bruises, sprains or strains

MTCs EXCLUDE

- First aid, LWCs and RWCs
- Hospitalisation for observation without treatment
- A one-off tetanus injection
- Consultative visit to, or examination by, a physician or registered professional for the purpose of a confirmatory check

What a First Aid Case includes

- Follow-up visits to a physician or nurse for observation only, or for a routine dressing change
- Negative X-ray results
- Cleaning abrasions and wounds with antiseptic and applying dressing
- Irrigation of the eye and removal of non-embedded foreign objects using a cotton swab
- One-time administration of oxygen after exposure to a toxic atmosphere, with resumption of normal — but not restricted — work the following day
- Soaking, application of a hot or cold compress and use of an elastic bandage on sprains and strains immediately after injury
- Applying a one-off cold compress or limited soaking of a bruise
- Use of non-prescriptive medicines and use of elastic bandages
- Treatment of first-degree burns

Exposure hours

Exposure hours are 24 hours per day while serving on board. This is the denominator for every frequency indicator, so it must be counted consistently across the fleet if the results are to be comparable.

The two frequency indicators

Lost Time Injuries are the sum of Fatalities, Permanent Total Disabilities, Permanent Partial Disabilities and Lost Workday Cases: LTIs = Fatalities + PTD + PPD + LWC. Total Recordable Cases are the sum of all work-related fatalities, lost time injuries, restricted work injuries and medical treatment injuries: TRCs = LTIs + RWCs + MTCs.

0.60

LTIF for three lost time injuries in 5,000,000 exposure hours

2.80

TRCF for 3 LWCs, 1 PTD, 1 PPD, 4 RWCs and 5 MTCs in 5,000,000 exposure hours

Lost Time Injury Frequency (LTIF) — the number of Lost Time Injuries per unit exposure hours. LTIF = LTIs x 1,000,000 / exposure hours. The most common unit is one million man-hours, so 3 x 1,000,000 / 5,000,000 = 0.60.

Total Recordable Case Frequency (TRCF) — the number of TRCs per unit exposure hours. TRCF = (LTIs + RWCs + MTCs) x 1,000,000 / exposure hours, so (3 + 1 + 1 + 4 + 5) x 1,000,000 / 5,000,000 = 2.80.

Exposure hours — 24 hours per day while serving on board, counted for all fleet personnel over the reporting period.

CLASSIFICATION

Decision tree for incident classification

First establish whether the accident is work related: an accident resulting in injury whilst assigned to a ship, off ship or on board, while engaged in non-work-related activity — personal, recreational or social — is a non-work-related accident; while engaged in work related activity it is a work related accident. Then classify the injury.

1. **Fatality?**
   If yes, the case is a Fatality. If no, continue.
2. **Permanent total or partial disability?**
   If yes, the case is a PTD or PPD. If no, continue.
3. **Work days lost?**
   If yes, the case is a Lost Time Injury. If no, continue.
4. **Restricted work or restricted duties?**
   If yes, the case is a Restricted Work Case. If no, continue.
5. **Medical treatment?**
   If yes, the case is a Medical Treatment Case. If no, continue.
6. **First aid?**
   If yes, the case is a First Aid Case.

Report it, investigate it, learn from it

Every near miss reported is an incident that did not have to happen. The purpose of an investigation is to find the facts that lead to corrective action — never to find fault.

Report promptly, by phone first

Initial report within 24 hours

Ask "if not, why not?" every time

Share the lessons across the fleet